TTL - transistor - transistor Logic

CMOS - complementary Metal - Oxide semiconductors

Boolean algebra order priority:

( )

AND

XOR

OR

High

Low

AND = ·

XOR = (+)

OR = +

De Morgan Law:

$\overline{a \cdot b} = \bar{a} + \bar{b}$

$\overline{a + b} = \bar{a} \cdot \bar{b}$

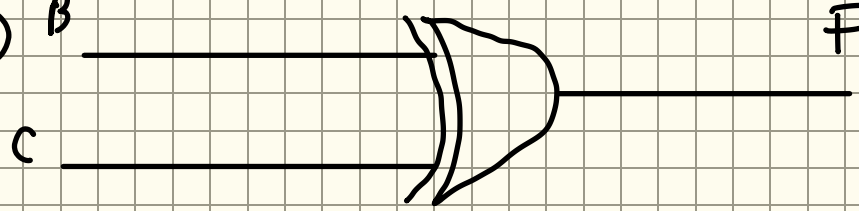
ex. 1) A



A	F_1
0	1
1	0

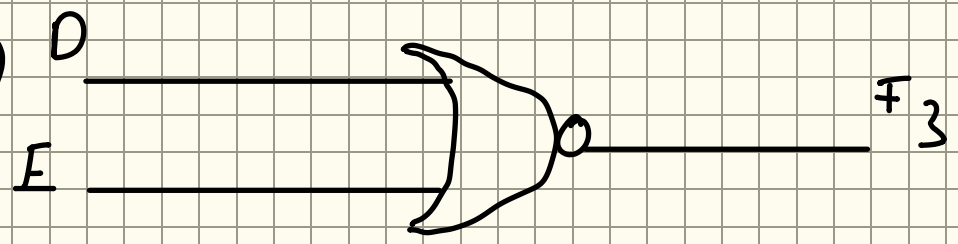
2) B

C



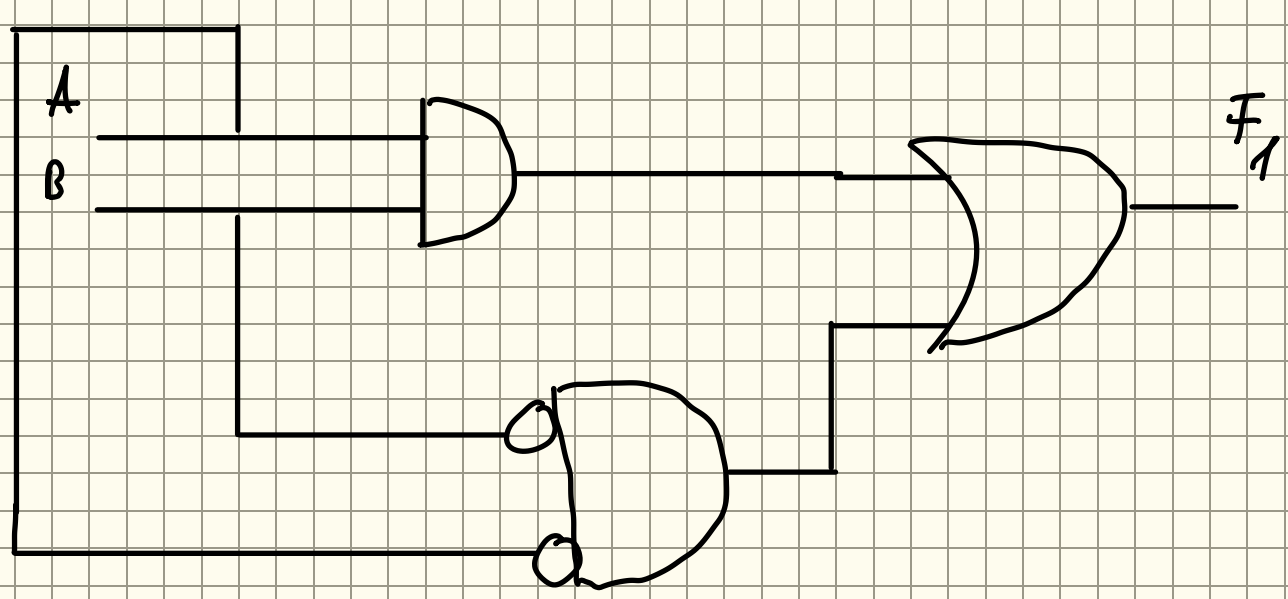
B	C	F_2
0	0	0
0	1	1
1	0	1
1	1	0

3)



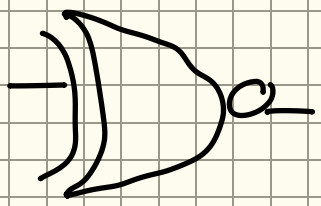
D	E	F_3
0	0	1
0	1	0
1	0	0
1	1	0

4)



A	B	F_1
0	0	1
1	0	0
0	1	0
1	1	1

F_1 = XNOR



5) $\bar{a} \cdot b + \bar{c} \Rightarrow$

$\bar{a}$ = 1 1 1 1 0 0 0 0

$\bar{c}$ = 1 0 1 0 1 0 1 0

a	b	c	F_1
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1

✓

1 1 1 | 0

6) $a \cdot c + \overline{b \cdot c} =$
$= a \cdot c + \bar{b} + \bar{c}$

a	b	c	F1
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

7) $a + \overline{b \cdot c}$

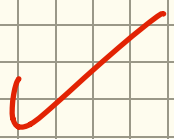
a	b	c	F1
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

8) $\overline{a + \bar{b}} \cdot c =$

a	b	c	F1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

9) $\overline{(a+b)\cdot\overline{a+c}}$

a	b	c	F_1
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



10) $a+\overline{b\cdot\overline{a+c}}$

a	b	c	F_1
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

